Detecting and Resolving CGM Absorption Components: S/N and Resolving-Power Requirements for Twelve Strong Transitions

Tanmay Singh

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Summary

- We compute, for H I λ1216, Mg II λ2796, Si II λ1260, S II λ1260, C II λ1335, Si III λ1206, Si IV λ1403, C IV λ1548, N V λ1239, O VI λ1032, Ne VIII λ770 and O VII λ21.6, the minimum column density detectable at 3σ and the smallest velocity separation at which two components can be told apart, as functions of S/N and resolving power R.
- Atomic data come from the Trident line list; Mg II λ2796 and O VII λ21.6, which it lacks, come from Morton (2003) and Verner et al. (1996).
- Both significances are linear in S/N, so each four-parameter problem is one three-dimensional cube times S/N.
- Classical resolution needs $\Delta v > \Delta v_{\rm S} = 0.85\,{\rm FWHM_{eff}}$ with ${\rm FWHM^2_{eff}} = (c/R)^2 + (1.665\,b)^2$; beyond $R_k = c/(1.665\,b)$ extra resolution gains little. Statistically, equal pairs follow $\Delta v_{\rm min} \simeq 1.98\,{\rm FWHM_{eff}}\,S_{\rm det}^{-1/4}$.
- Narrow low-ion components (Mg II, Si II, S II, C II, Si III; $b_{\rm fid} \approx 6\,{\rm km\,s^{-1}}$) gain a factor of ~ 2 in resolvable separation from $R = 17{,}000$ to 45,800; broad high-ion components (O VI, Ne VIII) gain only $\sim 25\%$ (Table 1).

1 Model and statistics

Each component has a Doppler optical-depth profile

$$\tau_i(v) = \tau_{0,i}\,e^{-(v-v_i)^2/b_i^2}, \qquad \tau_0 = \frac{\sqrt{\pi}e^2}{m_e c}\frac{N f \lambda}{b} = 1.497\times10^{-15}\,\frac{N[{\rm cm^{-2}}]\,f\,\lambda[\text{Å}]}{b[{\rm km\,s^{-1}}]}, \tag{1}$$

and the observed normalised flux is the convolution with the line-spread function (LSF), $F_{\rm obs} = \mathcal{L} \otimes e^{-\sum_i \tau_i}$. We take $\mathcal{L}$ to be a Gaussian with ${\rm FWHM_{LSF}} = c/R$ and $\sigma_{\rm LSF} = {\rm FWHM_{LSF}}/2.355$. Noise is constant at the continuum level, S/N is quoted per resolution element, and the spectrum is assumed Nyquist sampled.

Detection. For a known profile shape, the optimal (matched-filter) significance is

$$S_{\rm det} = {\rm S/N}\left[\frac{1}{\rm FWHM_{LSF}}\int D^2(v)\,dv\right]^{1/2}, \qquad D = 1 - F_{\rm obs}. \tag{2}$$

In the weak, unresolved limit this gives $W_{\lambda,3\sigma} \simeq 3.7\,\lambda/(R\,{\rm S/N})$.

Separation. Two components are *resolved* when the best-fitting single component, with free (N_1, b_1, v_1), is rejected. The expected penalty is $\Delta\chi^2 = S^2_{\rm res}$, with

$$S_{\rm res} = {\rm S/N}\left[\min_{N_1,b_1,v_1}\frac{1}{\rm FWHM_{LSF}}\int (F_2 - F_1)^2 dv\right]^{1/2}. \tag{3}$$

Here F_2 is the true two-component spectrum and F_1 the single-component model, both convolved with $\mathcal{L}$. We call a pair resolved at 3σ when $S_{\rm res} \geq 3$. A formal likelihood-ratio test with two or three extra free parameters at $p = 0.0027$ needs $\Delta\chi^2 \approx 11.8$–$14.2$, i.e. S/N about 15–25% higher.

Because Eqs. (2) and (3) are linear in S/N, every requirement follows from a single cube, $K(\log N, b, R)$ or $G(\log N, \Delta v, R)$. Then ${\rm S/N_{min}} = 3/K$ or $3/\sqrt{G}$, and contours of constant S/N are contours of the cube.

2 How the line-spread function limits component separation

Effective width. For optically thin components, each observed component is a Gaussian of dispersion $\sigma_{\rm eff}$, with

$$\sigma_{\rm eff}^2 = \frac{b^2}{2} + \sigma_{\rm LSF}^2, \qquad {\rm FWHM_{eff}} = \left[\left(\frac{c}{R}\right)^2 + (1.665\,b)^2\right]^{1/2}. \tag{4}$$

The LSF and the intrinsic width add in quadrature. Only $\rm FWHM_{eff}$ matters for blending.

Classical limit. Two equal Gaussians show a central dip only when their separation exceeds the Sparrow limit,

$$\Delta v_{\rm S} = 2\sigma_{\rm eff} = 0.849\,{\rm FWHM_{eff}}. \tag{5}$$

For $R \ll R_k \equiv c/(1.665\,b)$ this is LSF-limited, $\Delta v_{\rm S} \simeq 0.85\,c/R$. For $R \gg R_k$ it saturates at $\sqrt{2}\,b$ (Fig. 1b). With $b_{\rm fid} = 18.4\,{\rm km\,s^{-1}}$ (O VI), $R_k \approx 9{,}800$; with $b_{\rm fid} = 5.6\,{\rm km\,s^{-1}}$ (Mg II), $R_k \approx 32{,}000$.

Statistical limit. In the optically thin limit the ratio $\rho \equiv S_{\rm res}/S_{\rm det}$ depends only on $\delta \equiv \Delta v/\sigma_{\rm eff}$ and on the equivalent-width fraction q of the first component. Here $S_{\rm det}$ is the detection significance of the blend as a whole. The best-fitting single Gaussian absorbs the blend's area, centroid and second moment, so ρ is set by the first cumulant it cannot absorb:

$$q = \tfrac{1}{2}\ \text{(equal pair)}: \quad \kappa_3 = 0,\ \ \rho \simeq \frac{\sqrt{6}}{384}\,\delta^4 \ \Rightarrow\ \frac{\Delta v_{\rm min}}{\rm FWHM_{eff}} \simeq 1.98\,S_{\rm det}^{-1/4}, \tag{6}$$

$$q \neq \tfrac{1}{2}: \quad \kappa_3 = q(1-q)(1-2q)\,\Delta v^3,\ \ \rho \simeq \frac{q(1-q)|1-2q|}{4\sqrt{3}}\,\delta^3 \ \Rightarrow\ \Delta v_{\rm min} \propto S_{\rm det}^{-1/3}. \tag{7}$$

At large separations $\rho \to \min(q, 1-q)$: the pair can only be resolved if the weaker component is itself detected, $S_{\rm det} \geq 3/\min(q, 1-q)$. Equations (6)–(7) match the full numerical solution to better than 0.5%, both for the expansion coefficients at small δ and against the full numerical model at $\tau_0 \ll 1$.

For an equal pair the blend must be detected at $S_{\rm det} \approx 11$, 27, 43 and 270 to resolve separations of 1.5, 1.0, 0.85 (Sparrow) and 0.5 $\rm FWHM_{eff}$ (Fig. 1c). The steep $S_{\rm det}^{-1/4}$ law has a practical consequence. Below R_k, raising R is far cheaper than raising S/N. Above R_k, only S/N helps, at a cost of $\propto \Delta v_{\rm min}^{-8}$ in exposure time.

3 Twelve transitions

Each ion is given a fiducial Doppler parameter $b_{\rm fid} = (b_{\rm th}^2 + b_{\rm turb}^2)^{1/2}$, with the thermal width at a representative temperature (10^4 K for H I and the singly ionised species, rising to 10^6 K for O VII) and $b_{\rm turb} = 5\,{\rm km\,s^{-1}}$. Table 1 and Fig. 2 summarise the requirements; the per-ion instrument-plane and absorber-plane figures (`figures/ions/`) give the full maps. Three regularities stand out.

1. **Detection scales with $f\lambda$.** The minimum detectable column density tracks $\log N(\tau_0{=}1)$, set by $f\lambda/b$, to within a few tenths of a dex at fixed S/N and R, so the strongest transitions (Si II, Si III, Mg II) reach $\log N \approx 12$ at modest S/N.
2. **Resolution scales with b.** The resolvable separation depends on the ion almost entirely through $\rm FWHM_{eff}$. Low ions with $b \approx 6\,{\rm km\,s^{-1}}$ are LSF-limited up to $R \approx 30{,}000$; high ions are intrinsic-width-limited already at COS-like R.
3. **Saturation degrades resolution.** Above $\tau_0 \approx 3$ per component the resolvable separation rises again because saturated cores mimic a single broad component; the weaker doublet members (not used here) recover part of this.

4 Assumptions and caveats

- **Gaussian LSF.** The HST/COS LSF has extended wings; the tabulated LSF will raise $\Delta v_{\rm min}$ for COS-like set-ups.
- **Doppler cores only.** Damping wings are neglected (adequate below $\log N \approx 18$ for H I).
- **Equal components with $b = b_{\rm fid}$**; unequal pairs are treated in the thin limit (Fig. 1c).
- **Expected significances**, with the component count known to be one or two. Blind model selection needs higher thresholds.
- **O VII** is shown on the same R grid for comparison; current X-ray gratings have $R \lesssim 10^3$ at 21.6 Å.

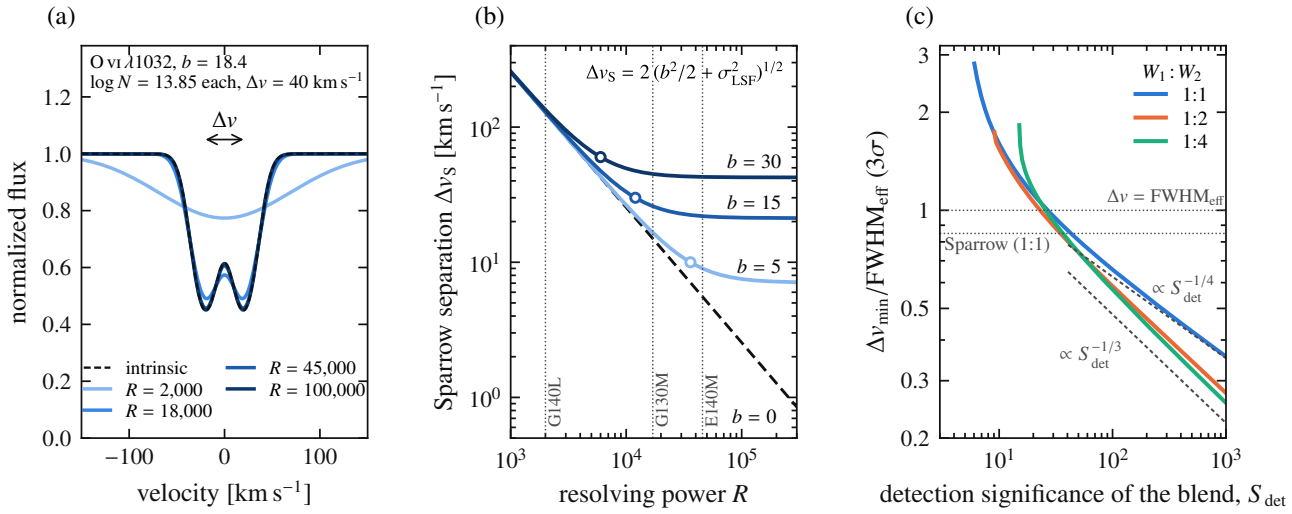


Figure 1: How the LSF limits component separation. (a) Two O VI components before and after convolution. (b) Sparrow separation versus R; circles mark R_k. (c) Universal 3σ resolution limit for optically thin pairs with equivalent-width ratios 1:1, 1:2 and 1:4.

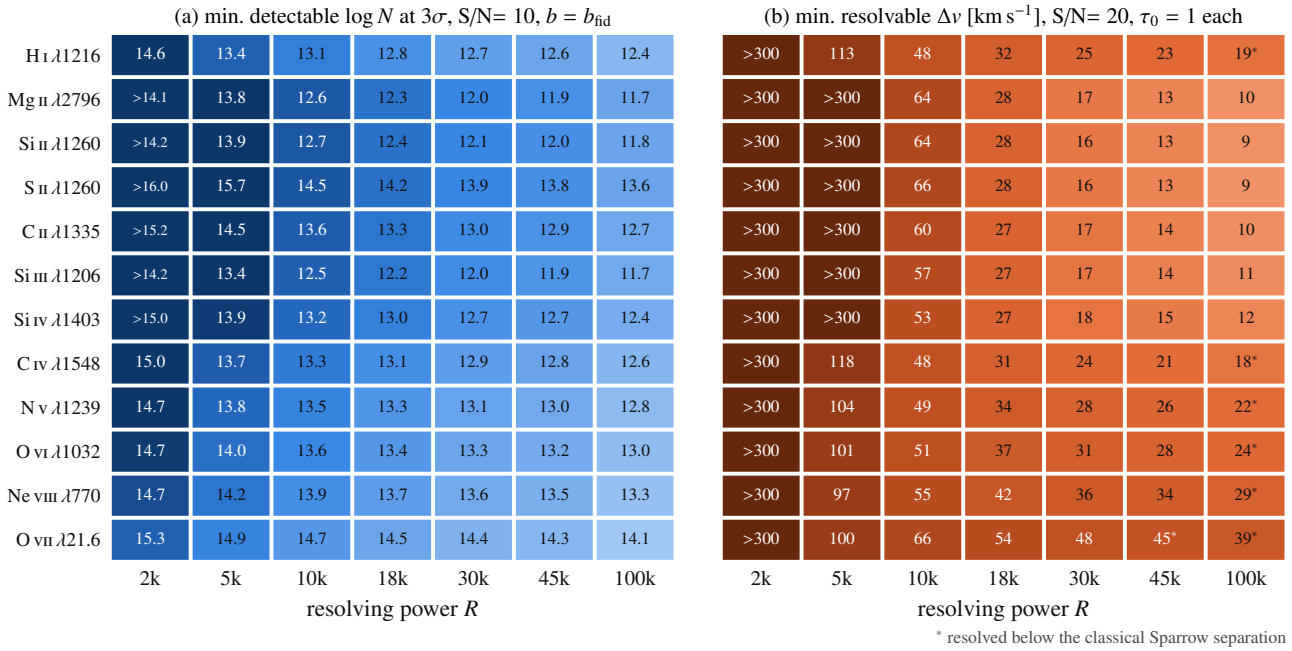


Figure 2: Requirement matrix for twelve transitions. (a) Minimum detectable log N at S/N = 10 and $b = b_{\rm fid}$. (b) Smallest resolvable separation for two equal $\tau_0 = 1$ components at S/N = 20.

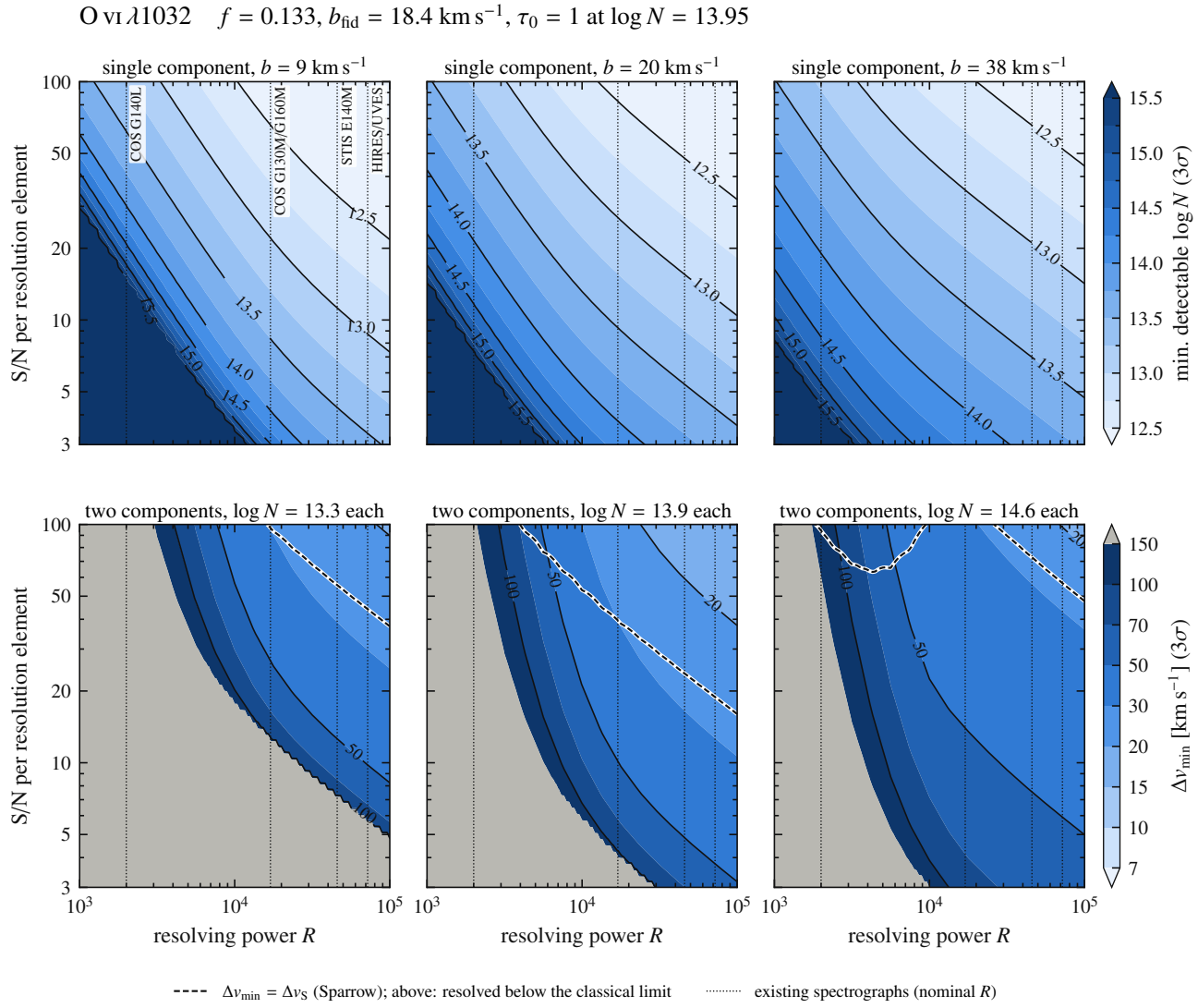


Figure 3: O VI λ1032 in the instrument plane. Top: minimum detectable log N for three b. Bottom: smallest resolvable separation for two equal components at three column densities. Inside the dashed contour, pairs are resolved below the Sparrow separation.

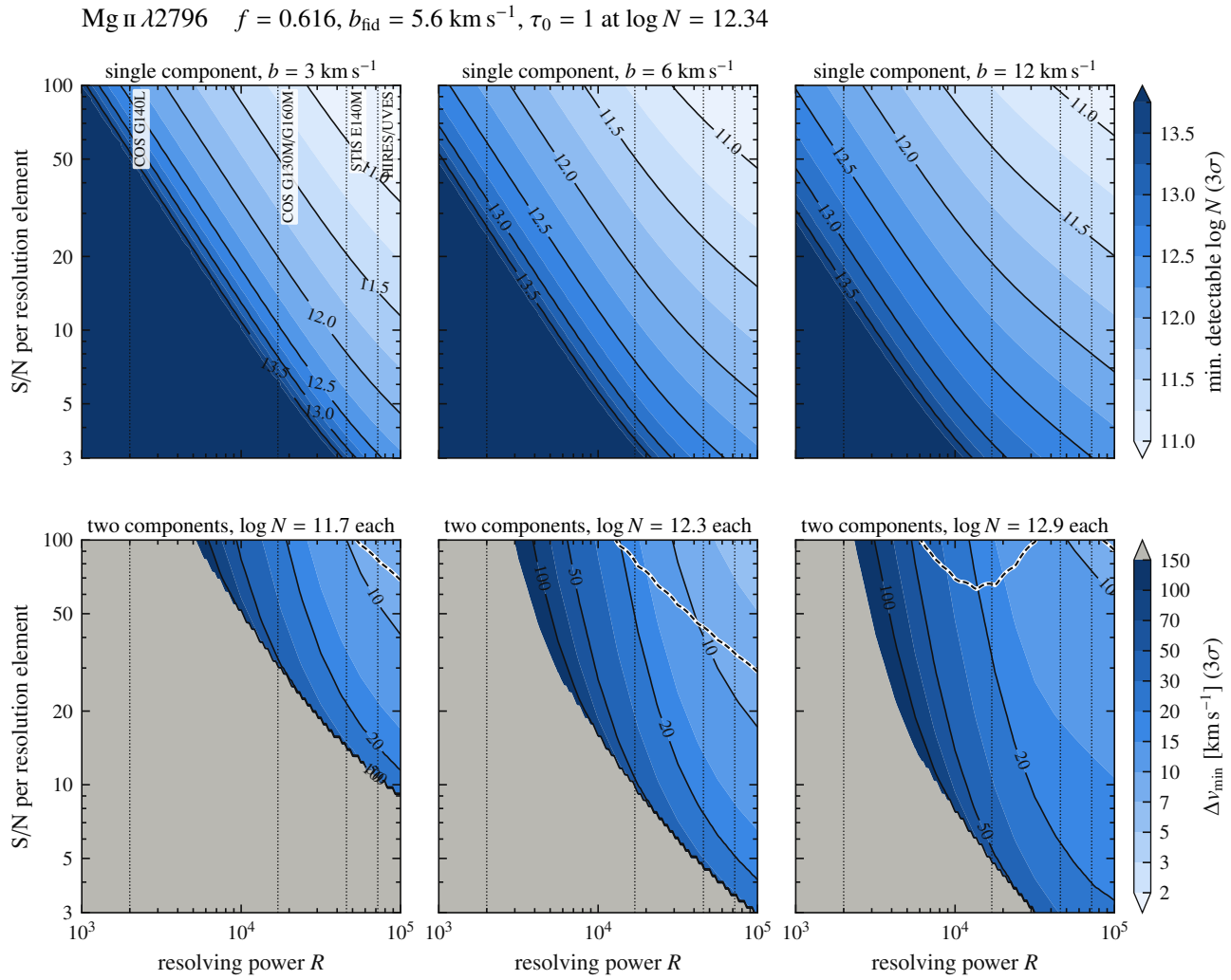


Figure 4: As Fig. 3 for Mg II λ2796.

Table 1: Requirements for twelve CGM transitions. Columns 5–7: minimum detectable $\log N$ (3σ, S/N = 10 per resolution element, $b = b_{\rm fid}$). Columns 8–10: smallest separation [km s^{-1}] at which two equal components, each with $\tau_0 = 1$, are resolved at 3σ with S/N = 20. *: below the classical Sparrow separation.

				$\log N_{\rm min}$			$\Delta v_{\rm min}$		
Transition	f	$b_{\rm fid}$	$\log N(\tau_0{=}1)$	17,000	45,800	100,000	17,000	45,800	100,000
H I λ1216	0.416	13.8	13.26	12.85	12.57	12.39	31.9	23.0	19.2*
Mg II λ2796	0.6155	5.6	12.34	12.29	11.89	11.69	27.6	13.3	9.5
Si II λ1260	1.18	5.6	12.40	12.35	11.95	11.75	27.6	13.2	9.4
S II λ1260	0.0182	5.5	14.20	14.17	13.76	13.56	27.8	13.2	9.3
C II λ1335	0.129	6.2	13.38	13.28	12.90	12.71	27.1	13.9	10.2
Si III λ1206	1.63	6.5	12.35	12.22	11.85	11.66	26.9	14.2	10.6
Si IV λ1403	0.255	7.8	13.16	12.96	12.62	12.43	27.2	15.6	12.1
C IV λ1548	0.19	12.8	13.46	13.07	12.79	12.61	30.9	21.8	18.0*
N V λ1239	0.156	16.2	13.75	13.28	13.01	12.84	34.5	26.0	21.9*
O VI λ1032	0.133	18.4	13.95	13.45	13.19	13.02	37.0	28.6	24.3*
Ne VIII λ770	0.102	22.8	14.29	13.72	13.47	13.31	42.3	33.9	29.1*
O VII λ21.6	0.696	32.6	15.16	14.50	14.26	14.10	54.5	45.4*	39.3*